Topology $(X, \mathcal{T})$

set X, collection of subset $\mathcal{T}$

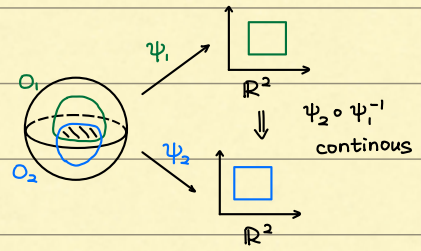
任意并，可数交封闭且 $X, \phi \in \mathcal{T}$

Smooth Manifold 局部可欧

M can be covered by $\cup O_i$ open set

$O_i \sim \mathbb{R}^d$ homeomorphic

$\psi_2 \circ \psi_1^{-1}$ smooth Atlas / Chart (O, ψ)



Smooth Manifold

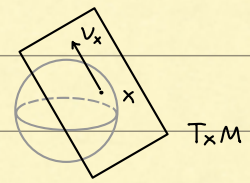
Tangent Space T_xM

Directional Derivative $Dh(x)[v] = \lim_{t \to \infty} \frac{1}{t}[f(x+tv) - f(x)]$

Given curve $\gamma(t)$

$v_x : \mathcal{F}_M \to \mathbb{R}$ M 上 vector 是泛函

$f \mapsto \frac{d}{dt} f \circ \gamma \big|_{t=0}$



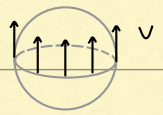
Tangent Space

T_xM tangent space

$TM = \bigcup_{x \in M} T_xM$ tangent bundle, $v_x \in T_xM \subset TM$

Vector Field $v : M \to TM$

$x \mapsto v_x$



Vector Field

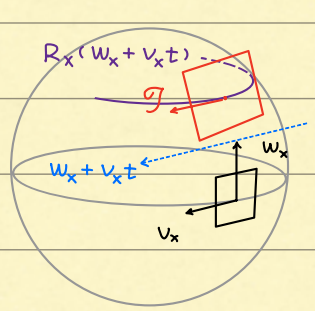
Vector Transport

$\mathcal{T} : TM \times TM \to TM$

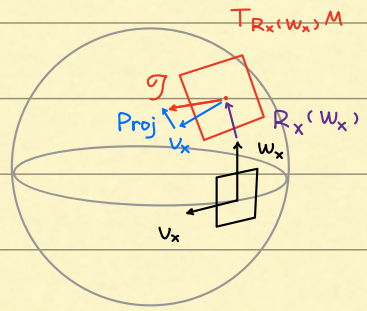
$(v_x, w_x) \mapsto \mathcal{T}(v_x, w_x)$ $\mathcal{T}_{v_x} w_x$

$\mathcal{T}(v_x, w_x) = \frac{d}{dt} R(w_x + v_x t)\big|_{t=0}$

$= \mathrm{Proj}_{T_{R_x(w_x)}M}(v_x)$



$\mathcal{T} = \frac{d}{dt} R_x(w_x + v_x t)\big|_{t=0}$



$\mathcal{T} = \mathrm{Proj}_{T_{R_x(w_x)}M}(v_x)$

Vector Transport

Stiefel Manifold

$St(n,p) = \{ X_{n\times p},\ X^TX = I_p \}$ $\quad n \geq p$

$h(X) = X^TX - I_p \qquad St(n,p) \to Sym(p)$

$Dh(X)[V] = X^TV + V^TX$

$N_xM = \{ \underset{n\times p}{X}\ \underset{p\times p}{S} : sym(S) = 0 \}$ $\qquad sym(A) = \frac{1}{2}(A + A^T)$

$T_xM = \{ \underset{n\times p}{X}\ \underset{p\times p}{A} + \underset{n\times(n-p)}{X_\perp}\ \underset{(n-p)\times p}{B} : skew(A) = 0 \}$ $\qquad skew(A) = \frac{1}{2}(A - A^T)$

$Proj_{N_xM}(U) = X \cdot sym(X^TU)$

$Proj_{T_xM}(U) = U - Proj_{N_xM}(U)$

$\qquad = (I - XX^T)U + X\, skew(X^TU)$

Riemannian Gradient $\quad \nabla_R f(X) = Proj_{T_xM}(\nabla f(X))$

Retraction $\quad R_X(\alpha \nabla_R f) = Qfactor(X + \alpha \nabla_R f)$

QR-decomp, then get Q

Vector Transport $\quad \mathcal{T}(\alpha \nabla_R f,\ \nabla_R f) = Proj_{T_{R_X(\alpha\nabla_R f)}M}(\nabla_R f)$

Quotient Manifold

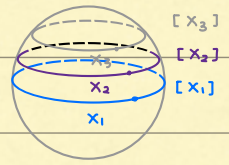
Quotient Manifold $\quad Q = M/\sim\ = \{ [X] \}$

Eqi-class $\quad [X] = \{ y \in M : y \sim X \}$

Canonial projection π : $\pi(X) = [X]$

Fiber $\mathcal{F} = \pi^{-1}(\pi(X))$ submanifold of M

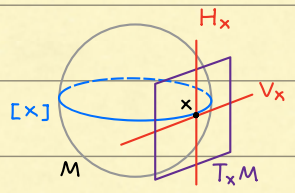
都一样



Quotient Manifold

Vertical Space $\quad V_xM = T_x\mathcal{F}$

Horizontal Space $\quad H_xM$, $\quad T_xM = V_xM \oplus H_xM$



$T_xM = V_x \oplus H_x$

Vertical / Horizontal Space

Lift : $T_{[X]}Q \to H_xM$

Lift 不用算，用于理解的

$[X] \in Q$

$u \in T_{[X]}Q$

$u = Lift_X(u)$

$= Proj_{H_X}(u) \in H_X$

Lift Operator

Riemannian Gradient

Given $f : [X] \to \mathbb{R}$, define $\bar{f}(X) = f([X])$

$$\nabla_R f([X]) = \mathrm{Lift}^{-1}[\mathrm{Proj}_{H_X M}(\nabla \bar{f}(X))]$$

Vector Transport

$$\mathcal{T}(W, V) = \mathrm{Lift}^{-1}[\mathrm{Proj}_{\pi^{-1}(H_{R_{[X]}(W)}M)}(\mathrm{Lift}(V))]$$

Grassman Manifold

$$Gr(n,p) = \{X\Theta : X \in St(n,p),\ \Theta_{p\times p} \in O(p)\}$$

$$M = St(n,p) \qquad Q = Gr(n,p) = St(n,p)/O(p)$$

$$[X] = \pi(X) = \mathcal{F} = \{X\Theta : \Theta_{p\times p} \in O(p)\}$$

$$H_X = \{X_\perp B\} \qquad V_X = \{XA : \mathrm{skew}(A) = 0\}$$

$$T_X M = \{X_{n\times p} A_{p\times p} + X_{\perp\, n\times(n-p)} B_{(n-p)\times p} : \mathrm{skew}(A) = 0\}$$

$$\mathrm{Proj}_{H_X}(U) = (I_n - XX^T)U$$

Riemannian Gradient

$$\nabla_R f([X]) = \mathrm{Lift}^{-1}[(I_n - XX^T)\nabla \bar{f}]$$ 算法里 Lift^{-1} 不用管

e.g. $f([X]) = \mathrm{tr}([X]^T A [X])$

$$\bar{f}(X) = \mathrm{tr}(X^T A X), \quad \nabla \bar{f} = 2AX$$

$$\nabla_R f = \mathrm{Lift}^{-1}[(I_n - XX^T)\cdot 2AX]$$

Retraction $[\hat{X}] = R_{[X]}(\nabla_R f) = [\mathrm{Qfactor}(X + \nabla_R f)]$

e.g. $X + (I_n - XX^T)\cdot 2AX$ 后 QF 分解，再取 $[\cdot]$

Vector Transport

$$\mathcal{T}(\alpha \nabla_R f, \nabla_R f) = \mathrm{Lift}^{-1}[(I - \hat{X}\hat{X}^T)\,\mathrm{Lift}\,\nabla_R f]$$

e.g. $\mathcal{T} = (I - \hat{X}\hat{X}^T)(I - XX^T)\cdot 2AX$